Introduction

Queue using a linked list
Data Structures and its Applications (UE25CS52A)

N S Kumar
Professor
Department of Computer Science & Engineering (AI & ML)

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Queue

1. On dequeue

a) Only head of the queue is changed

b) Only tail of the queue is changed

c) Both head and tail could change

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Queue

1. On dequeue

a) Only head of the queue is changed

b) Only tail of the queue is changed

c) Both head and tail could change

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Queue

2. To make both enqueue and dequeue efficient,

a) The pointer of each node will point to the node which joined the queue later

b) The pointer of each node will point to the node which joined the queue earlier

c) The pointer can point to either node before or node after

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Queue

2. To make both enqueue and dequeue efficient,

a) The pointer of each node will point to the node which joined the queue later

b) The pointer of each node will point to the node which joined the queue earlier

c) The pointer can point to either node before or node after

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Queue

3. Enqueue will change,

a) Only the head

b) Only the tail

c) Could change both the head and the tail

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Queue

3. Enqueue will change,

a) Only the head

b) Only the tail

c) Could change both the head and the tail

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Queue

4. If both the front and the rear are equal,

a) Implies that definitely the queue is empty

b) Implies that definitely the queue has one node

c) Implies either empty queue or one node queue

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Queue

4. If both the front and the rear are equal,

a) Implies that definitely the queue is empty

b) Implies that definitely the queue has one node

c) Implies either empty queue or one node queue

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Queue

5. Calling free on the front of a non empty queue,

a) Removes the first node and the rest of the queue is accessible

b) Remove the whole queue

c) Front automatically will point to the next node in the queue

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz : Queue

5. Calling free on the front of a non empty queue,

a) Removes the first node and the rest of the queue is accessible

b) Remove the whole queue

c) Front automatically will point to the next node in the queue

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Queue: interface

- Initialize / Deinitialize the queue
- enqueue
- dequeue
- Is_empty
- Is_full

© Confidential Information of PES University. All Rights Reserved - 2026

queue: structure

```
struct node
{
        int key_;
        struct node* next_;
};
typedef struct node node_t;
struct queue
{
        node_t* front_;
        node_t* rear_;
};
typedef struct queue queue_t;
```


© Confidential Information of PES University. All Rights Reserved - 2026

Queue: implementation

- Initialize / Deinitialize the queue
  - Initialize : make front and rear NULL
  - Deinitialize : deque all the nodes. make head and tail NULL
- Enqueue
  - Add at the rear
  - Modify rear
  - First node => modify front
- Deque
  - Remove the front
  - Modify front
  - Only node => modify rear

© Confidential Information of PES University. All Rights Reserved - 2026

Queue: Applications

- CPU scheduling : processes in batch processing usually wait in a queue
- Printer Spooler : collects the jobs for printing in a queue
- Breadth First search : Looking for a key in a tree structure
- Simulation models : queues to simulate random events in queueing theory
- Message queues : sharing information in operating systems
- Toll booth : waiting of vehicles to pass through a toll booth

© Confidential Information of PES University. All Rights Reserved - 2026

THANK YOU

N S Kumar
Professor

Department of Computer Science & Engineering (AI & ML)